MILITARY SPECIFICATION

SHIRTS, OPERATING, SURGICAL,
Man's, Cotton-Polyester, Static Resistant

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for one type and four sizes of a reversible, static resistant, cotton-polyester, man's, surgical, operating shirt.

1.2 Classification. The shirt shall be furnished in the following four sizes, as specified (see 6.2):

SIZES

Small	Medium	Large	Extra Large

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

DDD-L-20	Label, For Clothing, Equipage, and Tentage, (General Use)
PPP-B-585	Boxes, Wood, Wirebound
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	Boxes, Shipping, Fiberboard
PPP-B-676	Box, Setup

Military

MIL-L-10547	Liners, Case, and Sheet, Overwrap, Water-vaporproof or Waterproof, Flexible
MIL-T-43548	Thread, Polyester, Cotton-Covered
MIL-T-43624	Thread, Polyester, Spun

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:50Z
Check the source to verify that this is the current version before use.

STANDARDS

Federal

FED-Test Method Std No. 191	Textile Test Methods
FED-STD-751	Stitches, Seams and Stitchings

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

NATIONAL FIRE CODES

(Application for copies should be addressed to the National Fire Protection Association, International (NFPA), 470 Atlantic Avenue, Boston, Mass. 02110.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Materials. The materials shall conform to the applicable specifications and standards as listed or required herein.

3.1.1 Fabric

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:50Z
Check the source to verify that this is the current version before use.

3.1.1.1 Shirt. The shirts shall be fabricated from a fabric having a composition of a nominal 65 percent polyester, 34 percent combed cotton and one percent austenitic type, corrosion-resistant, stainless steel fibers having a diameter of 8 microns. The fabric shall conform to Table I when inspected as specified in 4.3.1 through 4.3.1.3, as applicable.

TABLE I. Cotton-Polyester Shirt Fabric Properties

Characteristics	Requirements
Weave	Plain
Ply of yarn:	
Warp	Single
Filling	Single
Yarns per inch, minimum:	
Warp	70
Filling	50
Breaking strength, pounds minimum:	
Warp	90
Filling	90
Ounces per square yard, minimum	4.8
Nonfibrous materials, percent maximum	2.0
pH	5.0 to 9.0
Shrinkage, percent maximum:	
Warp	3.0
Filling	3.0
Anti-Static properties, maximum:	
Surface resistivity, ohms per unit square	1×10^{11}
OR	
Static decay, seconds	0.5

3.1.1.1.1 Color. The color of the shirt fabric shall be a vat dyed Grayish Green shade. When inspected as specified in 4.3.1.2, the color of the cloth shall approximately match the color of the standard shade (see 6.3). The use of dyes or substances containing elementary sulfur or compounds capable of oxidation to sulfuric acid is prohibited. The dyed and finished fabric shall show not more than a slight trace of labile sulfur when inspected as specified in 4.3.1.

3.1.1.1.2 Colorfastness. When inspected as specified in 4.3.1, the dyed and finished fabric shall show fastness to light, laundering, perspiration and dry crocking equal to or better than the standard sample.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:50Z
Check the source to verify that this is the current version before use.

3.1.1.2 Pockets, binding, yoke and facing. The pockets, binding, yoke and facing shall be fabricated from the cotton-polyester-metallic fabric, 3.1.1.1 through 3.1.1.1.2.

3.1.3 Thread. All the sewing operations shall be performed with the cotton-covered, polyester thread conforming to MIL-T-43548, Ticket Number 70 and Ticket Number 50, 2 ply, or the spun polyester thread conforming to MIL-T-43624, Class 1 or 2, Ticket Number 70, 2 or 3 ply or Ticket Number 40, 3 ply and Ticket Number 50, 3 ply (see Table II). When spun polyester thread is used, the Ticket Number 40, 3 ply and Ticket Number 50, 3 ply may be substituted for the Ticket Number 50, 2 ply, and the Ticket Number 70, 2 or 3 ply may be substituted for the Ticket Number 70, 2 ply (see Table II). The color of the thread shall approximately match the color of the fabric, 3.1.1.1.1, when inspected as specified in 4.3.1.2.

3.2 Design. The shirt shall be a reversible, collarless, slip-over style with a continuous yoke back that terminates at the end of the cap sleeves and a total of two miter breast pockets so arranged that regardless which side is out, a breast pocket shall be on the left exterior side. The shirt shall conform in appearance to Figure 1.

3.3 Construction.

3.3.1 Cutting. The shirts shall be cut in strict accordance with the patterns furnished by the Government (see 6.4), which provide for seam allowance, size, shape, and placement of the pockets, directional lines for cutting, and notches for the proper assembling of all the parts. The Government patterns shall not be altered in any manner and shall be used for making the working patterns. The working patterns shall be identical to the Government furnished patterns. The component parts shall be cut from the same ply of the lay.

3.3.2 Shade and size marking. All the component parts of the shirts shall be marked or ticketed by any commercial method to insure a uniform shade and proper assembly throughout the shirt. No metal fastening device or sewn-on ticket shall be used. The shade and size markings shall be accomplished by the use of an ink pad numbering machine, rubber stamp, or pencil, provided the markings do not show through to the outerside of the shirt and are not deleterious to the cloth. The markings, which are placed on the seam allowance, shall not be visible on the outerside. The adhesive type shade and size marking tickets may be used for the shade and size markings, provided they conform to 3.3.2.1. The assembled shirt shall not contain any shade or size marking ticket nor any marking that was used for locating the position for attaching of any component

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:50Z
Check the source to verify that this is the current version before use.

3.3.2.1 Adhesive shade and size marking tickets. The adhesive shade and size marking tickets shall be fabricated from paper and shall contain a thermo-activated type adhesive on one side. The adhesive shall not discolor or damage the cloth and the adhesive mass shall not adhere to the cloth, upon removal of the ticket. The heat used to attach the thermoactivated adhesive tickets shall not stiffen, scorch, or damage the cloth in any manner.

3.3.3 Replacement of defective components and cut parts. Care shall be exercised during the spreading, cutting and manufacturing processes to assure that any component and cut part containing any material defect or damage that is classified in this specification as a defect shall be replaced with a non-defective and properly matched component or cut part, as applicable.

3.3.4 Seams and stitching. All the seams and stitching used in the fabrication of the shirts shall conform to Table II. The seam and stitch types shall conform to FED-STD-751. When two or more seam or stitch types are specified for the same part of the operation, the contractor may select one of them for the operation. The minimum and maximum number of stitches for each operation shall conform to Table II. When Stitch Type 401 is used, the looper thread shall be on the innerside of the garment. The two rows of stitching joining the sides, sleeves and yoke shall be 1/4 to 5/16 inch apart The outer row of the stitching shall be 1/16 inch from the edge. The seam allowances shall be maintained so that no runoff or open seam shall occur. All seams shall start and finish uniformly. No seam or component shall be twisted, puckered, or pleated and no part of the shirt shall be caught in an unrelated operation or seam. The neck facing and bindings shall be uniform in width, smooth and flat. All the thread ends shall be trimmed to a maximum length of 1/4 inch. No cut or raw edge shall be visible on the interior or exterior of the shirt.

3.3.4.1 Sewing. Each row of stitching shall be straight and parallel to the seam edge. The straightness of the stitching, in any row, shall be maintained within a tolerance of plus or minus 1/32 inch. The same tolerance for spacing and parallelism shall be applicable for the double rows of stitches. The ends of Stitch Type 301 stitching that are not caught in other seams or stitching shall be backstitched, by overlapping on themselves, a minimum of 1/2 inch. Each thread break, skip, and run-off shall be overstitched not less than 1/2 inch. Skipped stitches or thread breaks on Stitch Type 401 stitching may be repaired using Stitch Type 301. The thread tension shall be maintained so that there shall not be any loose or tight stitching and the lock shall be embedded in the material sewn together.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:50Z
Check the source to verify that this is the current version before use.

3.3.5 Manufacturing operation requirements . The shirts shall be manufactured in accordance with the operation requirements specified in Table II. The supplier is not required to follow the exact sequence of operations listed, unless otherwise specified. Any additional basting or holding stitches used to facilitate the fabrication of the shirts are permissable, provided the thread does not show on the surface of the assembled shirt.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:50Z
Check the source to verify that this is the current version before use.

TABLE II

NO	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM & STITCHING TYPE	STITCHES PER INCH	THREAD	
					NEEDLE	BOBBIN OR LOOPER
1.	Hem pockets					
	a. Turn the top of the pocket down with the raw edge turned under and stitch to the pocket 1/16 to 1/8 inch from the edge of the hem. The width of the hem shall be 1 ± 1/8 inch.	301	EFb-1	8 to 12	50/2	50/2
2	Set pockets					
	a. Turn under raw edges of the sides and bottom for the amount specified in the patterns and stitch the pocket to the left front breast. The stitching shall be 1/16 to 1/8 inch from the pockets edges	301	LSd-1	8 to 12	70/2	70/2
	b Turn over the front and sew on other pocket to the left front breast in the same manner as specified in 2a (see 3.2). The pockets shall be arranged so that regardless which side is out, the exterior left breast shall contain a pocket.	301	LSd-1	8 to 12	70/2	70/2
	c Each attached pocket shall be 4-3/4 ± 1/4 inches wide and 5 ± 1/4 inches long.					
3.	Attach the "V" neck binding					
	a. The cut edges of the "V" neck front shall be sewn within the folds of the binding The stitches shall be 1/16 to 1/8 inch from the edge. The binding shall overlap the "V" neck opening edges by 1/4 to 3/8 inch. The width of the binding on each side shall be 1-1/4 ± 1/8 inches. The right side of the binding shall overlap the left binding at the point of the "V". Stitch down "V" point portion with one row of horizontal stitching 1 to 1-1/4 inches from the point and also each side to form a triangle of stitching. The additional side stitching shall be 1 to 1-1/8 inches long.	301	BSc-1	8 to 12	50/2	50/2

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:50Z
Check the source to verify that this is the current version before use.

TABLE II (Cont'd)

NO	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM & STITCHING TYPE	STITCHES PER INCH	THREAD	
					NEEDLE	BOBBIN OR LOOPER
	b Reinforce the left side of the horizontal stitching at the "V" point with a 3/8 inch long horizontal bartack. The bartack shall be located 1/8 inch from the side of the binding.	Bartack		21	50/2	50/2
4.	Attach the neck facing to the yoke					
	a Superimpose the back neck facing or the yoke in the back neck area and attach together with one row of stitches 1/4 to 3/8 inch from the edges.	301	SSa-1	8 to 12	50/2	50/2
	b. Turn the facing onto the yoke and fold under the raw edges 1/4 to 3/8 inch, and attach to the yoke. The stitching shall be 1/16 to 1/8 inch from the edge The facing shall be on the outerside of the garment (see Figure 1)	301	LSd-1	8 to 12	50/2	50/2
5.	Join the side seams and sleeve extensions					
	a. Join each side seam and sleeve extension with a double lapped, double stitched seam. The seam may lap in either direction The stitching shall be 1/16 inch from the turned edge.	301 or 401	LSc-2	8 to 12	50/2	50/2
			LSc-2	8 to 12	50/2	70/2
6.	Join the shoulder seams (yoke and front) and sleeve extensions					
	a. Join each shoulder seam (yoke and front) and sleeve extension with a double lapped, double stitched seam The seam may lap in either direction The stitching shall be 1/16 inch from the turned edge.	301 or 401	LSc-2	8 to 12	50/2	50/2
			LSc-2	8 to 12	50/2	70/2
	b. Reinforce the shoulder seam, 1/4 inch from each neck opening, with a 3/8 inch long vertical bartack	Bartack		21	50/2	50/2

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:50Z
Check the source to verify that this is the current version before use.

TABLE II (Cont'd)

NO	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM & STITCHING TYPE	STITCHES PER INCH	THREAD	
					NEEDLE	BOBBIN OR LOOPER
7.	Join back, yoke and sleeve extensions					
	a. Join the back panel, yoke and sleeve extensions with a double lapped, double stitched seam The seam may lap in either direction The stitching shall be 1/16 inch from the turned edge.	301 or 401	LSc-2	8 to 12	50/2	50/2
			LSc-2	8 to 12	50/2	70/2
8.	Hem sleeve extensions					
	a. Fold under the end of each sleeve extension with the raw edge turned under to form a hem 1/2 ± 1/8 inch wide. The stitching shall be 1/16 to 1/8 inch from the edge.	301	EFb-1	8 to 12	50/2	50/2
9	Hem the bottom of the shirt					
	a. Fold under the bottom of the shirt with the raw edges turned under to form a hem 1/2 ± 1/8 inch wide. The stitching shall be 1/16 to 1/8 inch from the edge.	301	EFb-1	8 to 12	50/2	50/2
10.	Attach labels					
	a. Position the identification label on the center of the back, on the innerside of the garment, within 1/2 inch, with the top of the label 1/4 ± 1/8 inch below the top of the neck. Attach the label on all four sides.	301	LSbj-1	8 to 12	50/2	50/2
	b. Position the washing information label just below the identification label and attach all four sides.	301	LSbj-1	8 to 12	50/2	50/2
	Note. The top edge of the washing information label may be caught in the bottom seam of the identification label. Attach the remaining three sides.					
11.	Trim all thread ends and remove all thread scraps and foreign matter.					
12.	Fold each shirt to an approximate size of 14 by 12 inches without creating any wrinkles or creases.					

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:50Z
Check the source to verify that this is the current version before use.

3.4 Dimensions. The shirts after assembly shall conform to the dimensions for the different sizes specified in lable III when measured as specified in 4.3.4. The variation in length between the left and right sleeve dimensions shall not exceed 1/2 inch in any shirt.

TABLE III
SHIRT DIMENSIONS IN INCHES

Location	Size Small	Medium	Large	Extra Large	Tolerance
Chest	23-3/8	25-3/8	27-3/8	29-3/8	± 1/2
Sweep	49-1/4	52-1/4	56-1/4	60-1/4	± 1
Length	29-1/2	30	31	31-1/2	± 1/2

3.5 Labels.

3.5.1 Identification The identification label shall conform to DDD-L-20 Type I, Class 1, Launderable Label. The size of the label shall be adequate to contain the following information

SHIRT, OPERATING, SURGICAL, MAN'S
COTTON-POLYESTER, STATIC RESISTANT
MIL-S-37039
SIZE
CONTRACT OR ORDER NUMBER
NSN
WOOL OR FIBER PRODUCTS ACT INFORMATION

3.5.1.1 Alternate identification. An alternate identification label fabricated from cotton coutil or poplin cloth having a greige weight of 2.11 yards per pound (45 inch width) may be used in lieu of the label specified in 3.5.1. The printing medium and printing shall conform to DDD-L-20, Type I, Class 1, Launderable Label. The label shall contain the same information as specified in 3.5.1.

3.5.2 Washing information. The washing information label shall conform to DDD-L-20, Type I, Class 3, Launderable Label. The label shall contain the following information.

1. For maching washing, use wash and wear cycle or durable press cycle. Wash with moderately hot water (140°F), using soap or mild detergent.

2. For hand washing, wash in moderately hot water as described above. DO NOT WRING OR TWIST.

3. Dry in tumble dryer at medium setting.
Remove immediately at end of drying
Shape and hang.

4. To drip dry, remove from water before last spin cycle.
Shape and hang.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:50Z
Check the source to verify that this is the current version before use.

5. If touching up is required, iron at a moderate heat setting.

DO NOT REMOVE THIS LABEL

3.5.3 Washing instructions A lightweight, white paper washing instruction label with legible and durable black characters shall be placed on top of the folded shirt prior to packaging. The paper label shall be visible and capable of being read through the polyethylene film bag (see 5.1.3). The size of the letters and numerals shall be 10 point minimum in capitals. The size of the paper instruction label shall be at the option of the supplier. The label shall contain the following information:

WASHING INSTRUCTIONS

OPERATION	LEVEL (INCHES)	TEMPERATURE °F	TIME MINUTES	SUPPLIES PER 100 POUNDS OF GARMENTS
1. Flush	12	Cold	3	none
2. Suds	6	140	10	sodium metasilicate 1.5 pounds sodium tripolyphosphate (STPP) 0.5 pounds nonionic detergent 1.0 pounds (1 pint) brightener (whites only) 1.0 ounce
After 10 minutes at 140°F raise to 160°F for 10 minutes				
3. Rinse	10	150	3	
4. Bleach (white only)	10	150	5	1% sodium hypochlorite 2.0 quarts (rinse colors instead)
5. Rinse	10	135	3	
6. Rinse	10	120	3	
7. Rinse	10	105	3	
8. Sour	6	Cold	5	ammonium silicofluoride 1.0 ounce

Notes a. Wash Polyester-Cotton only with Polyester-Cotton
b. Wash whites separately Be sure wheel is clear
c. Load wheel only to 2/3 its rated capacity

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:50Z
Check the source to verify that this is the current version before use.

3.6 Workmanship. The shirts shall not contain any spot, stain, hole, abraded area, tear, cut, mend patch, scorch, or needle chew. The shirts shall conform to the quality of grade established by this specification. The occurrence of defects shall not exceed the acceptance criteria established herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Material and component inspection. In accordance with 4.1, the materials and components shall be inspected and tested in accordance with all the requirements of the referenced specifications and standards, unless otherwise excluded, amended, modified, or qualified in this specification or purchase document.

4.1.2 Certificate of quality. When certificates of quality are submitted in accordance with Table IV, they shall contain verifiable actual test and inspection data. The Government reserves the right to inspect and test the components to verify the validity of the certification.

4.1.3 Inspection. Inspection, as used in this specification, is defined as both examination (an investigation without the use of special laboratory apparatus or procedures such as visual) and testing (a determination by technical means of the physical and chemical properties) of the item.

4.2 Quality conformance inspection. The sampling and inspection shall conform to MIL-STD-105, except where otherwise indicated herein after. The quality conformance inspection shall consist of the tests and examinations specified in Table IV.

4.2.1 Sampling.

4.2.1.1 Inspection lot.

4.2.1.1.1 Shirt fabric. An inspection lot size shall be expressed in units of one linear yard of the shirt fabric and shall consist of all the shirt fabric received by the shirt manufacturer at one time. The sample unit for inspection shall be two linear yards.

4.2.1.1.2 Alternate identification label fabric. An inspection lot size shall be expressed in units of one linear yard of the alternate identification label fabric and shall consist of all the alternate identification label fabric received by the shirt manufacturer at one time. The sample unit for inspection shall be one linear yard.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:50Z
Check the source to verify that this is the current version before use.

4.2.1.1.3 Shirts. An inspection lot size shall be expressed in units of one shirt made essentially under the same conditions and from the same materials and components. The sample unit for inspection shall be one shirt.

4.2.1.1.4 Preparation of delivery. An inspection lot size shall be expressed in units of one fully prepared shipping container, containing shirts of one size, fully prepared for delivery from essentially the same materials and components. The sample unit for inspection shall be one shipping container, containing shirts of the size, fully prepared for delivery with the exception that it need not be sealed.

4.2.1.2 Sampling for tests and examinations of the shirt and alternate identification label fabrics, shirts and preparation for delivery. The sample size, acceptance criteria, tests and examinations required for the shirt and alternate identification label fabrics, shirts, or the preparation for delivery, as applicable, shall be as specified in Table IV.

TABLE IV
SAMPLE SIZE, ACCEPTANCE CRITERIA, TESTS AND EXAMINATIONS OF THE SHIRT AND ALTERNATE IDENTIFICATION LABEL FABRICS, SHIRTS AND PREPARATION FOR DELIVERY

Item	Inspection	Paragraph		Sample	Acceptance	
		Requirement	Method	Size	Criteria	
Shirt Fabric 1/	Polyester Identification	3.1.1.1	4.3.1.1	2/	2/	3/
	Cotton Identification	3.1.1.1	4.3.1	2/	2/	3/
	Stainless steel Identification	3.1.1.1	4.3.1.1	2/	2/	3/
	Percent Polyester	3.1.1.1	4.3.1.1	2/	2/	3/
	Percent Cotton	3.1.1.1	4.3.1.1	2/	2/	3/
	Percent Stainless Steel Fiber	3.1.1.1	4.3.1.1	2/	2/	3/
	Diameter of Stainless Steel Fiber	3.1.1.1	4.3.1.1	2/	2/	3/
	Weave	3.1.1.1	4.3.1.1	2/	2/	3/
	Ply of Yarn	3.1.1.1	4.3.1.1	2/	2/	3/
	Yarns per inch	3.1.1.1	4.3.1	2/	2/	
	Breaking Strength	3.1.1.1	4.3.1	2/	2/	
	Weight	3.1.1.1	4.3.1	2/	2/	
	Non Fibrous Materials	3.1.1.1	4.3.1	2/	2/	

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:50Z
Check the source to verify that this is the current version before use.

TABLE IV (Cont'd)

Item	Inspection	Paragraph Requirement	Paragraph Method	Sample Size	Acceptance Criteria	
	pH	3.1.1.1	4.3.1	2/	2/	
	Shrinkage	3.1.1.1	4.3.1	2/	2/	
	Anti-Static Properties	3.1.1.1	4.3.1.3	2/	2/	
	Color	3.1.1.1.1	4.3.1.2	2/	2/	3/
	Kind of Dye	3.1.1.1.1	4.3.1.1	2/	2/	3/
	Labile Sulfur	3.1.1.1.1	4.3.1	2/	2/	3/
	Colorfastness to:					
	Light	3.1.1.1.2	4.3.1	2/	2/	
	Laundering	3.1.1.1.2	4.3.1	2/	2/	
	Perspiration	3.1.1.1.2	4.3.1	2/	2/	
	Dry Crocking	3.1.1.1.2	4.3.1	2/	2/	
Alternate Identification Label Fabric 4/ Shirts	Cotton Identification	3.5.1.1	4.3.2	2/	2/	
	Weight	3.5.1.1	4.3.3	2/	2/	
	Kind of Fabric	3.5.1.1	4.3.2.1			
	Visual Examination	3.3 through 3.3.5 and 3.5 through 3.6	4.3.3.1	Inspection level II	An acceptable quality level of 2.5 defects per 100 units for Major defects and 10.0 defects per 100 units for total (major plus minor) defects	
	Dimensional check	3.4	4.3.3.1	Inspection level S-2	An acceptable quality level of 4.0 defects per 100 units	
	Preparation for Delivery	Section 5	4.3.3.2	Inspection level S-2	An acceptable quality level of 2.5 defects per 100 units	

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:50Z
Check the source to verify that this is the current version before use.

1/ The inspection of the shirt and alternate identification label fabrics, as applicable, shall be conducted either at the contractor's or subcontractor's plant.

2/ The sampling for the testing of the shirt or alternate identification label fabric shall be in accordance with 4.2.1.1.1 and 4.2.1.1.2, as applicable, and as follows:

Lot size (yards)	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

The samples shall be taken from the beginning of each roll. The lot shall be unacceptable if one or more sample units fail to meet any of the requirements specified in Table IV. Each sample unit taken for inspection shall be used to determine all the characteristics of the shirt or alternate identification label fabric, as applicable, specified in Table IV.

3/ A certificate of quality shall be submitted and will be acceptable for the stated requirements (see 4.1.2).

4.3 Inspection methods.

4.3.1 Shirt fabric. The shirt fabric characteristics shall be determined in accordance with the inspection methods of FED-STD-191 specified in Table V and 4.3.1.1 through 4.3.1.3.

TABLE V
SHIRT FABRIC PROPERTIES

Characteristic	FED-STD-191 Method
Cotton identification	1200
Yarns per inch	5050
Breaking strength	5100
Weight	5041
Nonfibrous materials	2611
pH	2811
Shrinkage	5552
Labile sulfur	2020
Colorfastness to:	
Laundering (3 cycles)	5610 1/
Light	5660
Perspiration	5680
Dry crocking	5651

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:50Z
Check the source to verify that this is the current version before use.

1/ The staining of the cotton fiber on the color transfer cloth shall only be evaluated. The test specimen shall be dried after each laundering.

4.3.1.1 Polyester and stainless steel fiber, fiber content, weave, ply of yarn and kind of dye. The identification of the polyester and stainless steel fibers, fiber content, weave, ply of yarn and the kind of dye shall be determined by any standard commercial method.

4.3.1.2 Color matching. The color of the shirt fabric and the thread shall be compared to the approved shade under natural (north sky) daylight or artificial daylight having a color temperature of 7500 degrees Kelvin.

4.3.1.3 Anti-static properties. The anti-static properties of the shirt fabric shall be determined in accordance with NFPA-56A

4.3.2 Alternate identification label fabric. The characteristics of the alternate identification label fabric shall be determined in accordance with the inspection methods of FED-STD-191 specified in Table VI and 4.3.2.1.

TABLE VI
ALTERNATE IDENTIFICATION LABEL FABRIC

Characteristic	FED-STD-191 Method
Cotton identification	1200
Weight	5041

4.3.2.1 Kind of alternate identification label fabric The kind of fabric used for the alternate identification label shall be determined by any standard commercial method.

4.3.3 Visual examination.

4.3.3.1 Shirts. Each shirt selected as a sample unit from the lot shall be thoroughly checked dimensionally and examined visually to determine conformance to this specification. The classification and list of defects, Tables VII and VIII, as applicable, shall be used to classify and enumerate the defects found.

TABLE VII
CLASSIFICATION OF DEFECTS FOR THE VISUAL EXAMINATION
OF THE ASSEMBLED SHIRTS

DEFECT	MAJOR	MINOR
GENERAL		
a. Any hole, scissors or knife cut, tear, mend, burn, or weakening defect such as smash, multiple float, loose slub, needle chew, or abraded area		X

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:50Z
Check the source to verify that this is the current version before use.

TABLE VII (Cont'd)

DEFECT	MAJOR	MINOR
GENERAL (Cont'd)		
b. Any portion of the shirt stiffened or scorched by any process of manufacturing	X	
c. Any missing or broken yarn, thin place, misweave, woven in waste, loose yarn, snagged thread, area of no dye penetration, or dye streak on the exterior of the shirt	1/	
SHADED PARTS		
a. Any part shaded or any shade bar		X
b. Color of any component not as specified or not uniform		X
CLEANNESS		
a Three or more thread ends not trimmed to a maximum length of 1/4 inch or three or more thread scraps not removed		X
b. Any unsightly slub, spot, or stain; any discoloration of the cloth caused by the removal of any shade and size marking ticket; any shade and size marking ticket, excluding the metal fastening or sewn-on kind, not removed, any shade and size marking identification stamping visible on the outerside of the shirt, or any shade and size marking adhesive mass adhering to the cloth		X
c. Any metal fastening device or sewn-on shade and size marking ticket attached to the shirt	X	
d Any marking used for locating the position for attaching of any component visible on the top or underside of the shirt		X

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:50Z
Check the source to verify that this is the current version before use.

TABLE VII (Cont'd)

DEFECT	MAJOR	MINOR
COMPONENTS AND ASSEMBLY		
a. Any component not as specified or any defect of component or defect of assembly, not herein classified	1/	
b. Any component, component part, or required operation omitted, or any operation improperly performed, not herein classified	1/	
c. Any dimension not as specified, not herein classified	1/	
SEAMS AND STITCHING		
a. Accuracy of seaming:		
1. Any seam, hem, or attachment of any component twisted, puckered, or pleated, any part of the shirt or of any component caught in an unrelated operation or stitching; any seam edge not properly forced out or contains any fold		X
2. Ends of stitching that are not caught in other seams or stitching:		
(a) Any not securely backstitched	X	
(b) Any secure but backstitched for a distance less than 1/2 inch or the backstitching is not overlapped on itself		X
3. Thread breaks.		
(a) Any not securely overstitched	X	
(b) Any secure but overstitched for a distance less than 1/2 inch		X

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:50Z
Check the source to verify that this is the current version before use.

TABLE VII (Cont'd)

DEFECT	MAJOR	MINOR
SEAMS AND STITCHING (Cont'd)		
b. Gage of stitching 2/:		
1. Any stitching irregular, unevenly or not gaged as specified, or not uniform		X
c. Open seams 3/.		
1. Other than on the hem:		
(a) Any more than 1/8 inch but not more than 1/4 inch		X
(b) Any more than 1/4 inch	X	
2. On the hem:		
(a) Any more than 1/4 inch but not more than 1/2 inch		X
(b) Any more than 1/2 inch	X	
d. Raw edges 4/:		
1. Outerside or innerside of the shirt:		
(a) Any more than 1/4 inch but not more than 1/2 inch		X
(b) Any more than 1/2 inch	X	
e. Seam and stitch type		
1. Not specified seam or stitch	1/	
2. Any looper thread on the outerside of any shirt seam, when Stitch Type 401 is used	X	

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:50Z
Check the source to verify that this is the current version before use.

TABLE VII (Cont'd)

DEFECT	MAJOR	MINOR
SEAMS AND STITCHING (Cont'd)		
f. Stitch tension:		
1. Any loose stitching 5/	1/	
2. Any tight stitching, resulting in breakage of the stitches, when normal pull is applied 6/	X	
g. Stitches per inch 7/:		
1. Stitches more or less than specified:		
(a) Shirt (excluding labels):		
(1) One or 2		X
(2) More than 2	X	
(b) Labels		
(1) One to 3		X
(2) More than 3	X	
h. Bartacks:		
1. Any missing, loose, misplaced or not serving the intended purpose	X	
2. Length of any not as specified	1/	
3. Any not located as specified	1/	

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:50Z
Check the source to verify that this is the current version before use.

TABLE VII (Cont'd)

DEFECT	MAJOR	MINOR
DETAILED DEFECTS APPLICABLE TO SPECIFIC PARTS OF THE SHIRT		
FACING		
a. Short, tight, or twisted causing puckering or to the yoke	X	
BINDING		
a. Any cut on a bias	1/	
b. Any short, tight, or twisted causing puckering or twisting to the front	X	
POCKETS		
a. Any pocket out of horizontal or vertical alignment:		
1. More than 3/8 inch but not more than 1/2 inch		X
2. More than 1/2 inch		
b. Any not of uniform size and shape		X
c. The width of any pocket less than 4-1/2 inches or more than 5 inches		X
d. The length of any pocket less than 4-3/4 inches or more than 5-1/4 inches		X
e. Any corner not mitered		X
LABELS		
a. Identification:		
1. Size missing, incorrect, or illegible	X	
2. Other than the size information, incomplete, incorrect, or illegible		X
3. Misplaced, off center of the back by more than 1/2 inch		X
4. Stitching omitted on one or more edges		X

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:50Z
Check the source to verify that this is the current version before use.

TABLE VII (Cont'd)

DEFECT	MAJOR	MINOR
b. Washing information:		
1. Any washing information, missing incorrect, incomplete, or illegible	X	
2. Stitching omitted on one or more edges		X
3. Not located as specified		X
c. Washing instructions:		
1. Any washing instructions missing, incorrect, incomplete, or illegible	X	
2. Size of markings less than 10 points		X

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:50Z
Check the source to verify that this is the current version before use.

1/ The defect shall be classified and scored as a major defect, when it seriously affects the serviceability or appearance, otherwise it shall be classified and scored as a minor defect.

2/ The gage of stitching defect shall be scored only when the condition exists for more than 2 inches on any row of stitching.

3/ A seam shall be classified and scored as open when one or more stitches joining a seam are broken or when two or more continuous skipped stitches or run-offs occur. On double stitched seams, a seam shall be classified and scored as open when either one or both sides of the seam contain any of the aforementioned for a single stitched seam.

4/ Any edge that should be turned under and is not, but is securely caught in the stitching, shall be classified and scored as raw edge. Any raw edge, that is not securely caught in the stitching, shall be classified and scored as an open seam.

5/ A seam shall be classified and scored as loose when the top and bobbin threads are on the same surface.

6/ Puckering is evidence of tight tension or gathering of the material. When puckering is evident and is not caused by the gathering of the material, the stitching shall be inspected for tight tension and breaking of stitches by exerting normal pull in the lengthwise direction of the stitching, by pulling the material taut to straighten out the seam. Puckering due to the gathering of the material shall be classified and scored as an accuracy of seaming defect.

7/ The stitches per inch shall be scored only when the condition exists for more than 3 inches on any row of stitching.

TABLE VIII

LIST OF DEFECTS FOR THE DIMENSIONS OF THE ASSEMBLED SHIRTS

ASSEMBLED MEASUREMENT	DEFECT
Measure the chest, sweep and length	Any measurement deviating from the measurements specified 3.4 and Table III and applicable tolerance shall be enumerated as a dimensional defect.
Equal Length	
Sleeves	The variation in length between the left and right sleeve dimension shall be enumerated as a dimensional defect when it exceeds 1/2 inch.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:50Z
Check the source to verify that this is the current version before use.

4.3.3.2 Preparation for delivery. Each of the fully prepared shipping containers, containing shirts of one size, selected as a sample unit from the lot, shall be visually examined to determine that the packaging, packing, and marking conform to this specification. The list of defects, Table IX, shall be used to enumerate the defects.

TABLE IX
LIST OF DEFECTS FOR PREPARATION FOR DELIVERY

ITEM	DEFECT
Exterior and interior markings	Missing, incorrect, incomplete, or illegible; of improper size, location, sequence, or method of application; size shown on any exterior container or on the label not the same; information on the washing instruction paper label in the polyethylene bag not readable through the film or no label in the polyethylene bag.
Materials	Any non-conforming component, any component or component part missing, damage, or otherwise defective.
Workmanship	Inadequate application of the components such as incomplete closure of any exterior container, container flap, or loose strapping; bulging or distortion of any container, any polyethylene bag damaged, any vent hole missing, or any open or non-continuous heat sealed seam.
Exterior and interior weight or content	Gross or net weight exceeds the requirement, number per container more or less than specified; more than one size shirt in the same container, any portion of the shirt caught in any heat sealed seam, any shirt distorted, deformed, or bent, any not individually packaged within the polyethylene bag.

4.3.4 Procedure for measuring. The shirt to be measured shall be laid flat, without tension, on a smooth, flat surface so that any wrinkles or creases shall not affect the measurement. The measurements shall be made with the use of a yard stick, metal tape, or by use of any suitable measuring device. The measurements shall be made as follows:

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:50Z
Check the source to verify that this is the current version before use.

a. The chest dimension shall be taken at the base of the armholes, from side seam to side seam at a point 11 inches below the center of the neck.

b. The sweep dimension shall be taken along the entire bottom edge of the shirt.

c. The length dimension shall be taken along the center of the front from the top of the neck opening to the bottom edge of the shirt.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2)

5.1.1 Unit of issue. One package containing twelve shirts, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in table X shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table X Packaging quantities

Unit package	Intermediate package	Exterior container
1 unit	Not required	4 units

5.1.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units to be overpacked in an exterior container, such units may be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Unit package Each shirt shall be neatly folded to an approximate dimension of 14 inches in length and 12 inches in width and packaged in a clear, snug fitting polyethylene bag having a minimum thickness of 0.00125 inch. A 1/4 inch diameter hole shall be made at one corner of the bag to allow excess air to escape and the closure shall be effected by heat sealing or by means of a tuck or reverse flap Twelve shirts of the same size individually packaged in polyethylene bags shall be packaged in a snug fitting box constructed in accordance with PPP-B-636, type CF, class domestic or PPP-B-676 except that commercial colors are acceptable. Closure of the box shall be secured to prevent accidental spilling of contents under normal handling.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:50Z
Check the source to verify that this is the current version before use.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant. Fiberboard boxes shall conform to the special requirements specified in PPP-B-636.

5.2.1.2 Case liner. Each wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraph of appendix thereto. Case liner shall not be required for fiberboard boxes.

5.2.1.3 Closure. Closure of each wood box shall be in accordance with appendix of applicable box specification. Closure of each fiberboard box shall be as specified in the appendix and waterproofed in accordance with paragraph 30.4 of PPP-B-636.

5.2.1.4 Strapping. Strapping, when required, shall be in accordance with appendix of applicable box specification.

NOTE: Strapping shall not be required for shipment forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Fiberboard boxes shall conform to the special requirements as specified in PPP-B-636. Closure of wood boxes shall be as specified in the appendix of the applicable box specification. Closure of fiberboard boxes shall conform to method II as specified in PPP-B-636.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers at the lowest rate, to point of delivery called for in the contract or purchase order.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:50Z
Check the source to verify that this is the current version before use.

5.3 Marking.

5.3.1 Unit package. Each unit package shall be marked in accordance with MIL-STD-129. Contract or purchase order number, size, and name of contractor shall be shown. When the contractor is not the manufacturer, the name of the manufacturer shall also be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

NOTE. The National Stock Number shall be shown in lieu of the Federal Stock Number wherever specified in MIL-STD-129. Marking shall not be required on polyethylene bags.

5.3.3 Labels, mixed sizes. Each shipping container packed with mixed sizes shall have securely attached to the end and side, directly under the printing or stenciling, a white paper label 5 by 4 inches with the words "MIXED SIZES" plainly stamped or printed thereon and under these words shall be legibly stamped or printed the correct quantity and sizes contained therein.

6. NOTES

6.1 Intended use. The static resistant, cotton-polyester, man's, surgical, operating shirts covered by this specification are intended for wear by the male surgical personnel of the Medical Services.

6.2 Ordering data. Procurement documents shall specify the following.

a. Title, number, and date of this specification.

b. Size and quantity desired (see 1.2).

c. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

d. Whether any special markings are required (see 5.3).

6.3 Standard sample for shade and colorfastness. The standard sample for shade and colorfastness will be furnished by the contracting officer.

6.4 Patterns. The patterns for the shirts will be furnished by the contracting officer to the contractor for use in cutting working patterns (see 3.3.1). The pattern parts shall consist of the following:

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:50Z
Check the source to verify that this is the current version before use.

NOMENCLATURE	CUT PARTS
Front	One
Back	One
Yoke	One
Patch Pocket	2
Yoke facing	One
Front Neck Binding	2

6.5 Catalog Information. This specification is intended to cover the following items appearing in the Federal Supply Catalog.

NATIONAL STOCK NO.	ITEM IDENTIFICATION
6532-00-149-0322	SHIRT, OPERATING, SURGICAL, Man's Cotton-Polyester, Static Resistant Small, 12's
6532-00-149-0323	SHIRT, OPERATING, SURGICAL, Man's Cotton-Polyester, Static Resistant Medium, 12's
6532-00-149-0324	SHIRT, OPERATING, SURGICAL, Man's Cotton-Polyester, Static Resistant Large, 12's
6532-00-149-0325	SHIRT, OPERATING, SURGICAL, Man's Cotton-Polyester, Static Resistant Extra-Large, 12's

6.6 Commercial Information. This specification does not include all types, classes, styles or sizes of the commodity indicated by the title of this specification, or which are commercially available, but is intended to cover the items, which are normally procured to meet military requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

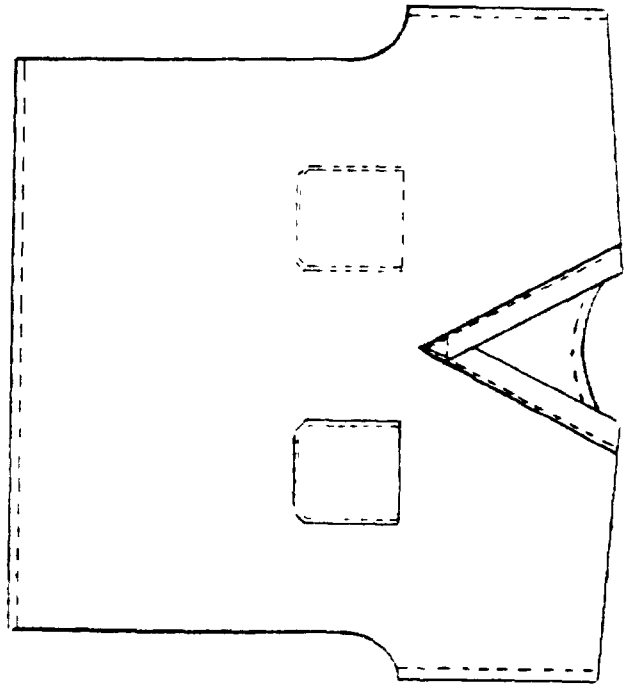
Review activities:
Army - MD
Navy - MS
Air Force - 03

Preparing activity.
Defense Supply Agency - DM

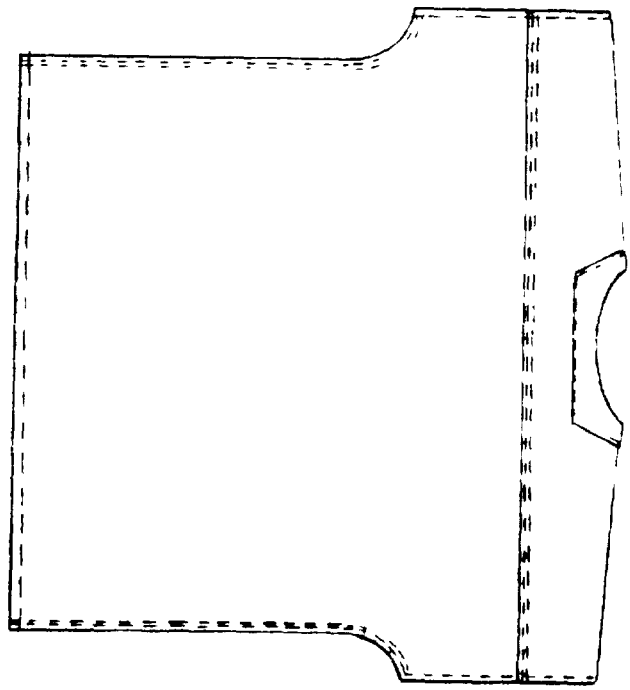
Project No. 6532-1075

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:50Z
Check the source to verify that this is the current version before use.



FRONT



BACK

FIGURE 1 SHIRT, OPERATING SURGICAL

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:50Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No 22-R255

INSTRUCTIONS The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document DoD contractors government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government Fold on lines on reverse side staple in corner, and send to preparing activity Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements Attach any pertinent data which may be of use in improving this document If there are additional papers attach to form and place both in an envelope addressed to preparing activity

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3 IS THE DOCUMENT RESTRICTIVE

☐ YES ☐ NO *(If "Yes" in what way?)*

4 REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED S/N 0102-014-1802

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:50Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

HEADQUARTERS, DEFENSE PERSONNEL SUPPORT CENTER
ATTN DIRECTORATE OF MEDICAL MATERIEL, CODE ATT
2800 SOUTH 20TH STREET
PHILADELPHIA, PA 19101

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:50Z
Check the source to verify that this is the current version before use.